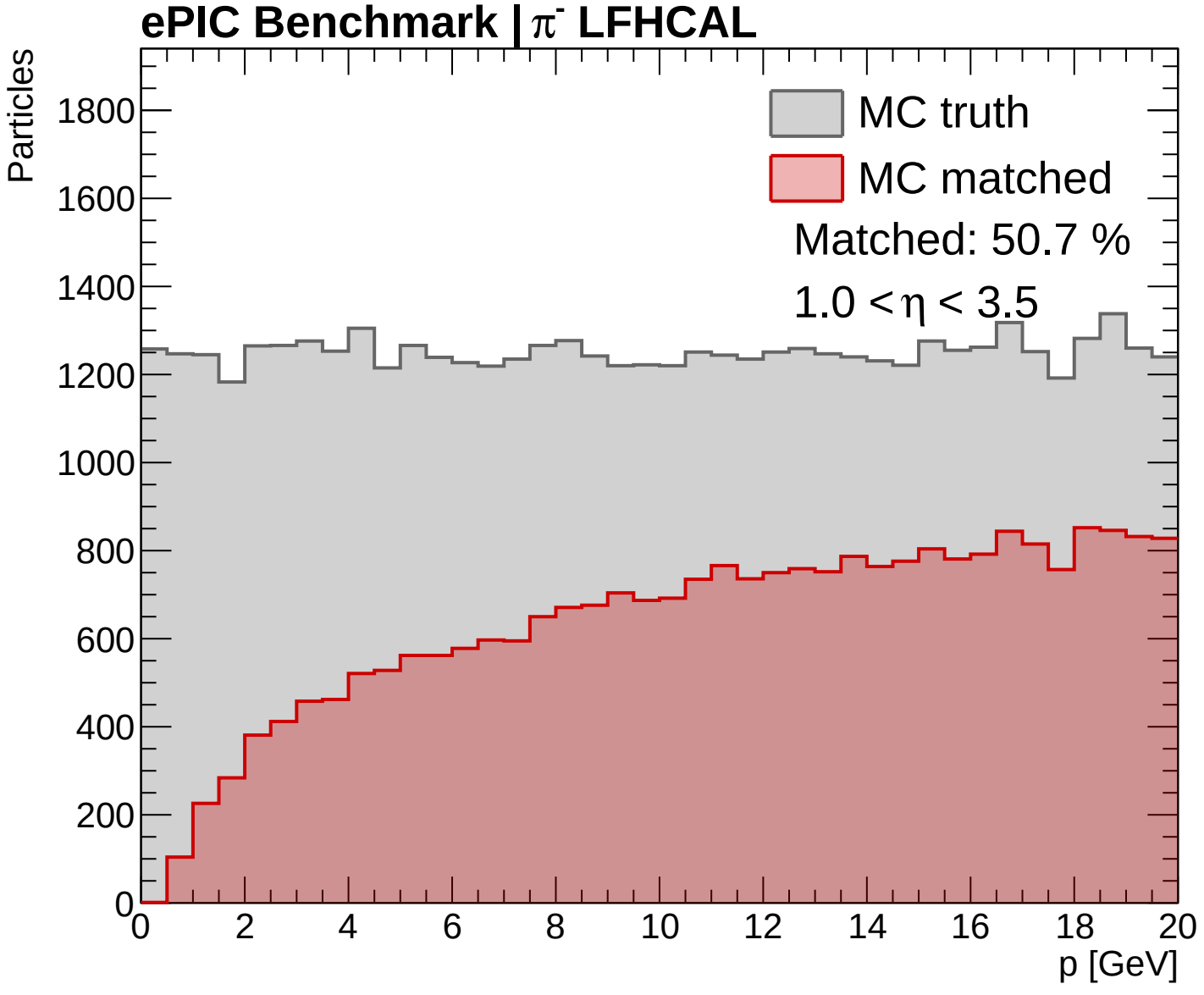
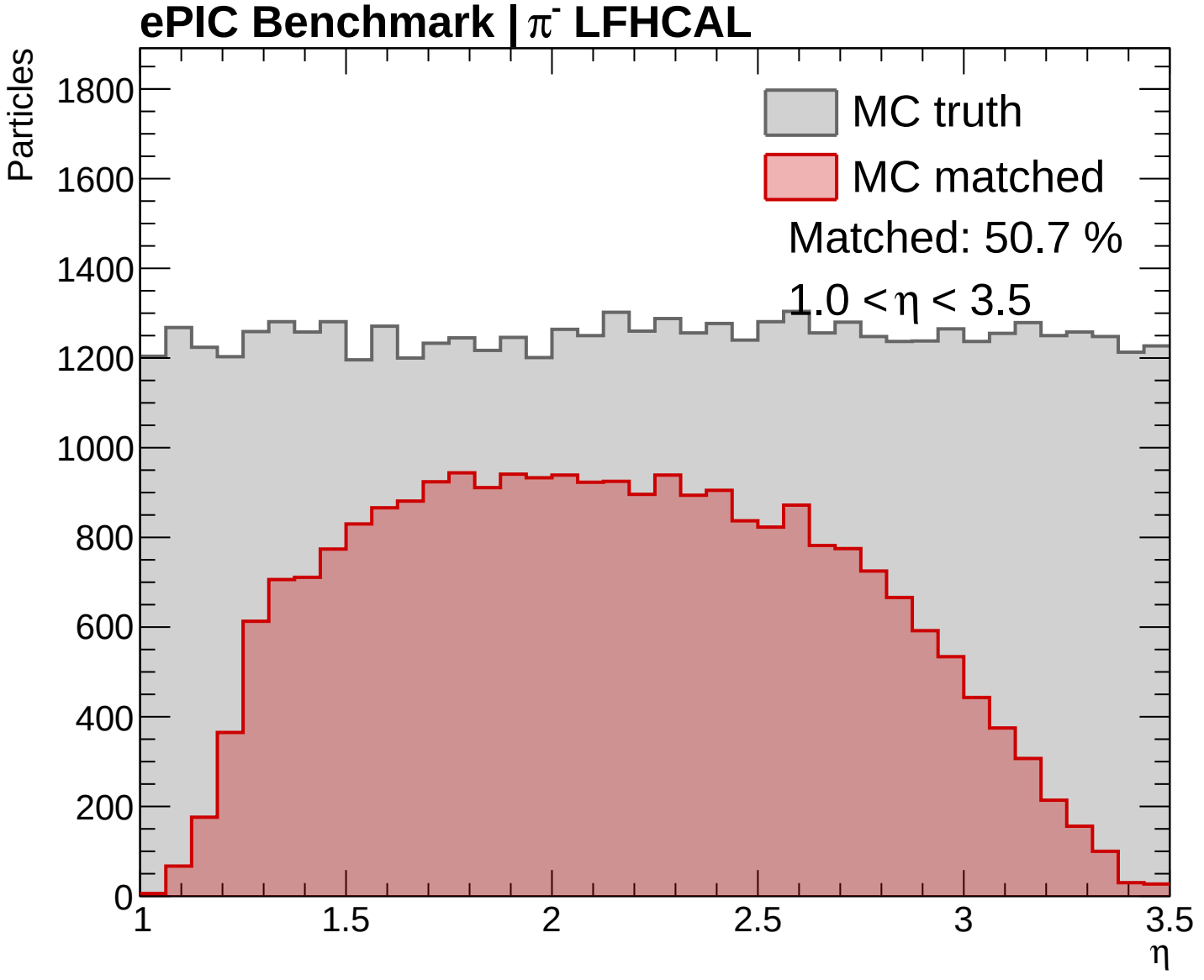
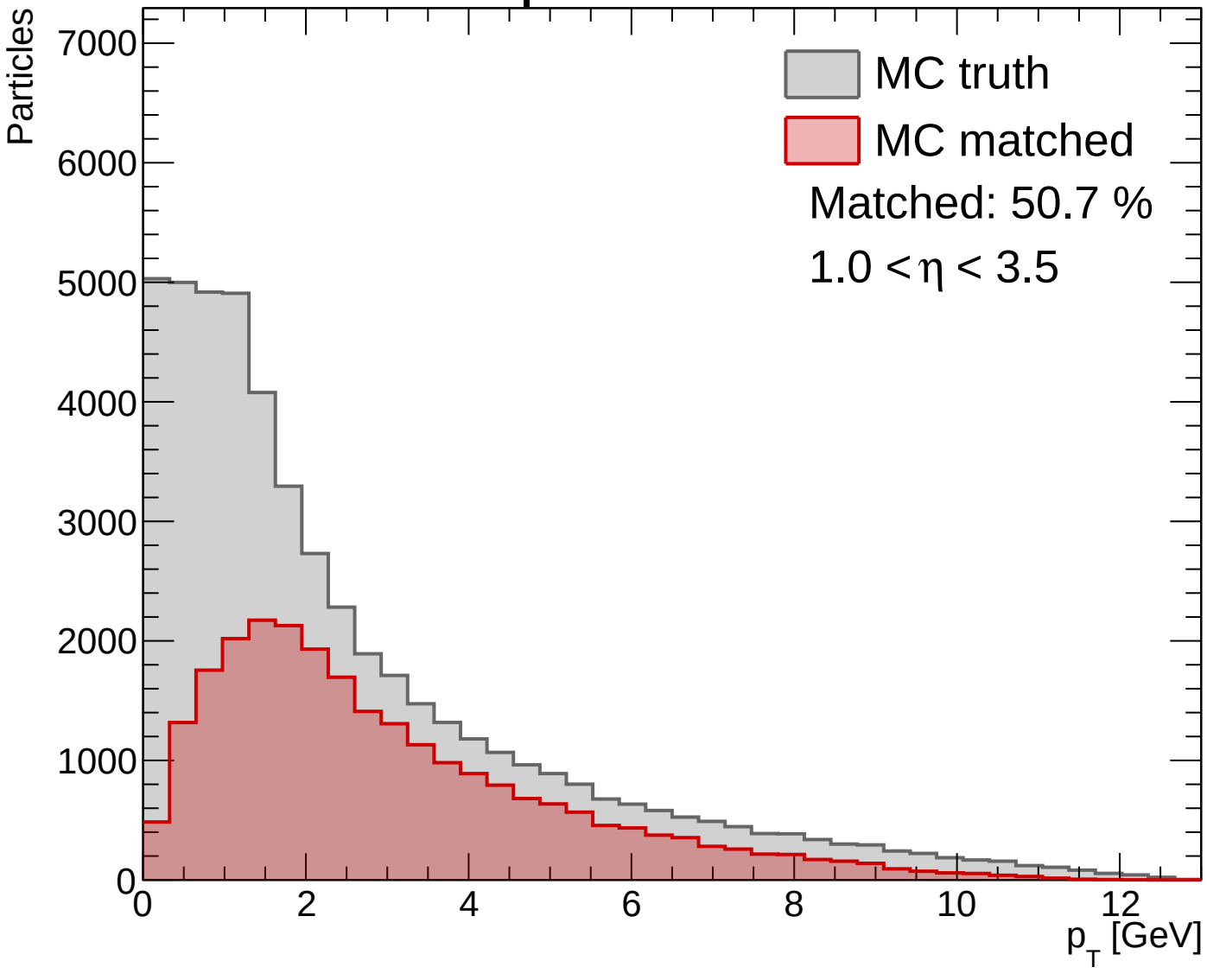
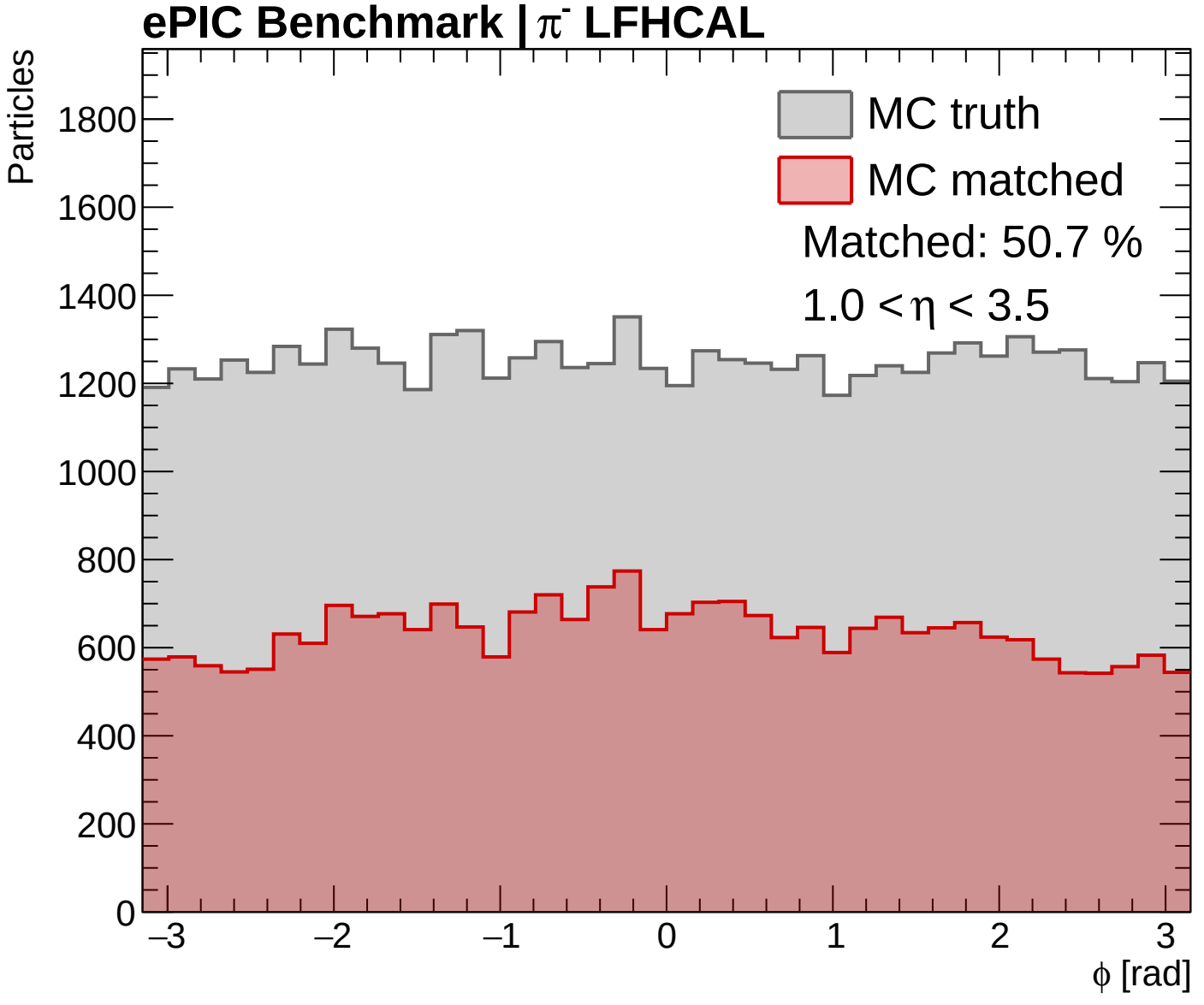
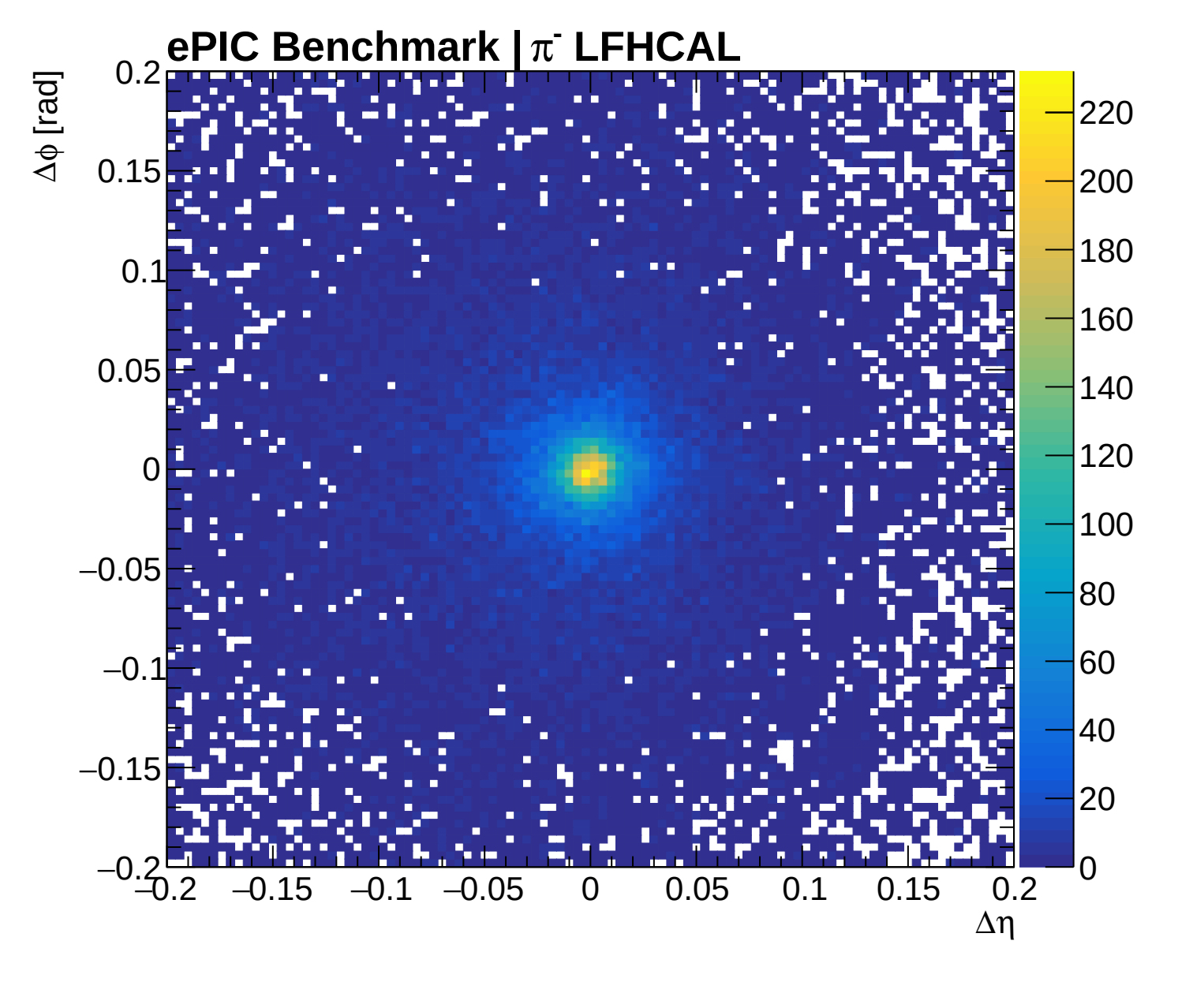
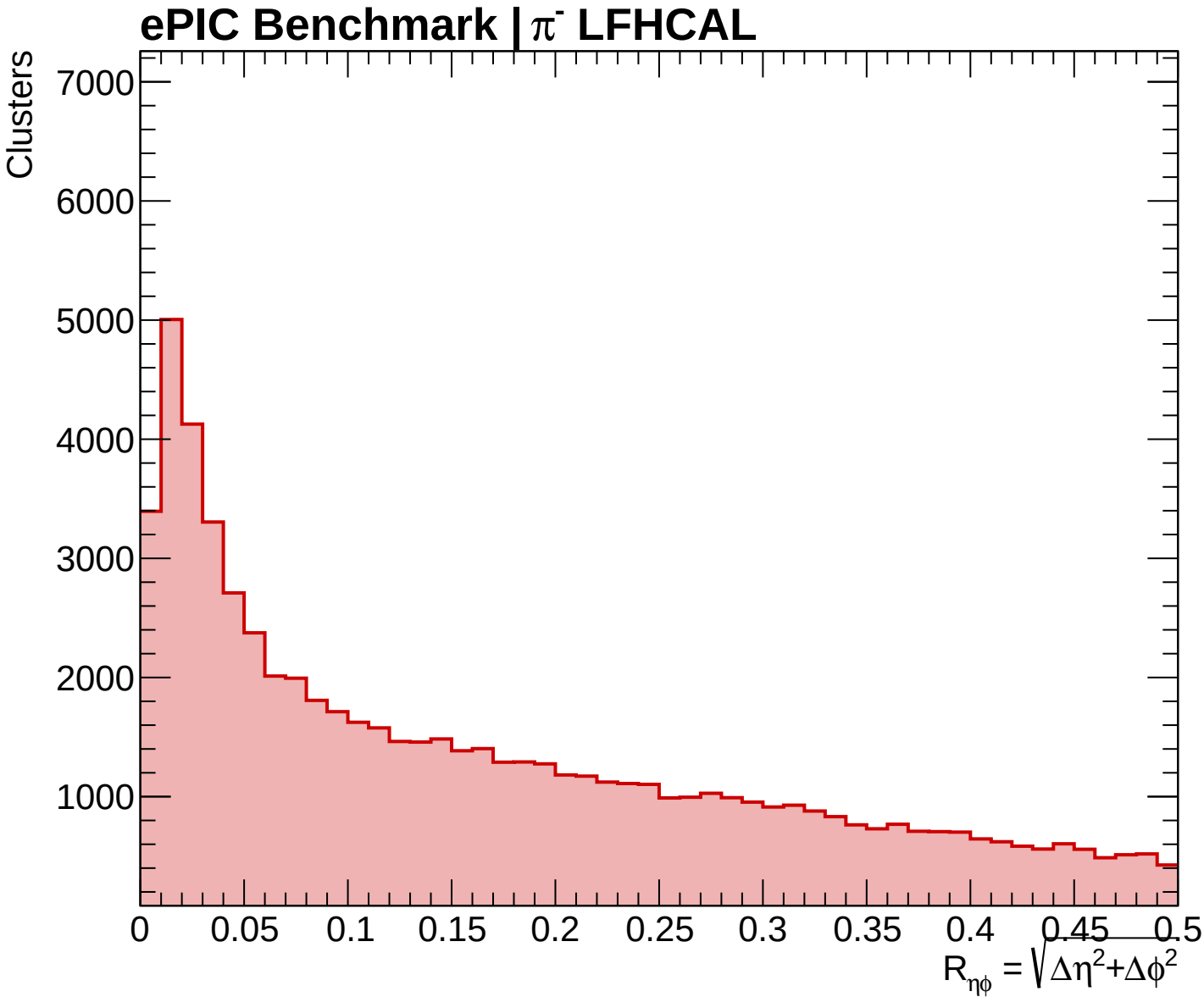

ePIC Benchmark | π⁻ LFHCAL
Particles
MC truth
MC matched
Matched: 50.7 %
1.0 < η < 3.5
0
200
400
600
800
1000
1200
1400
1600
1800
0
2
4
6
8
10
12
14
16
18
20
p [GeV]

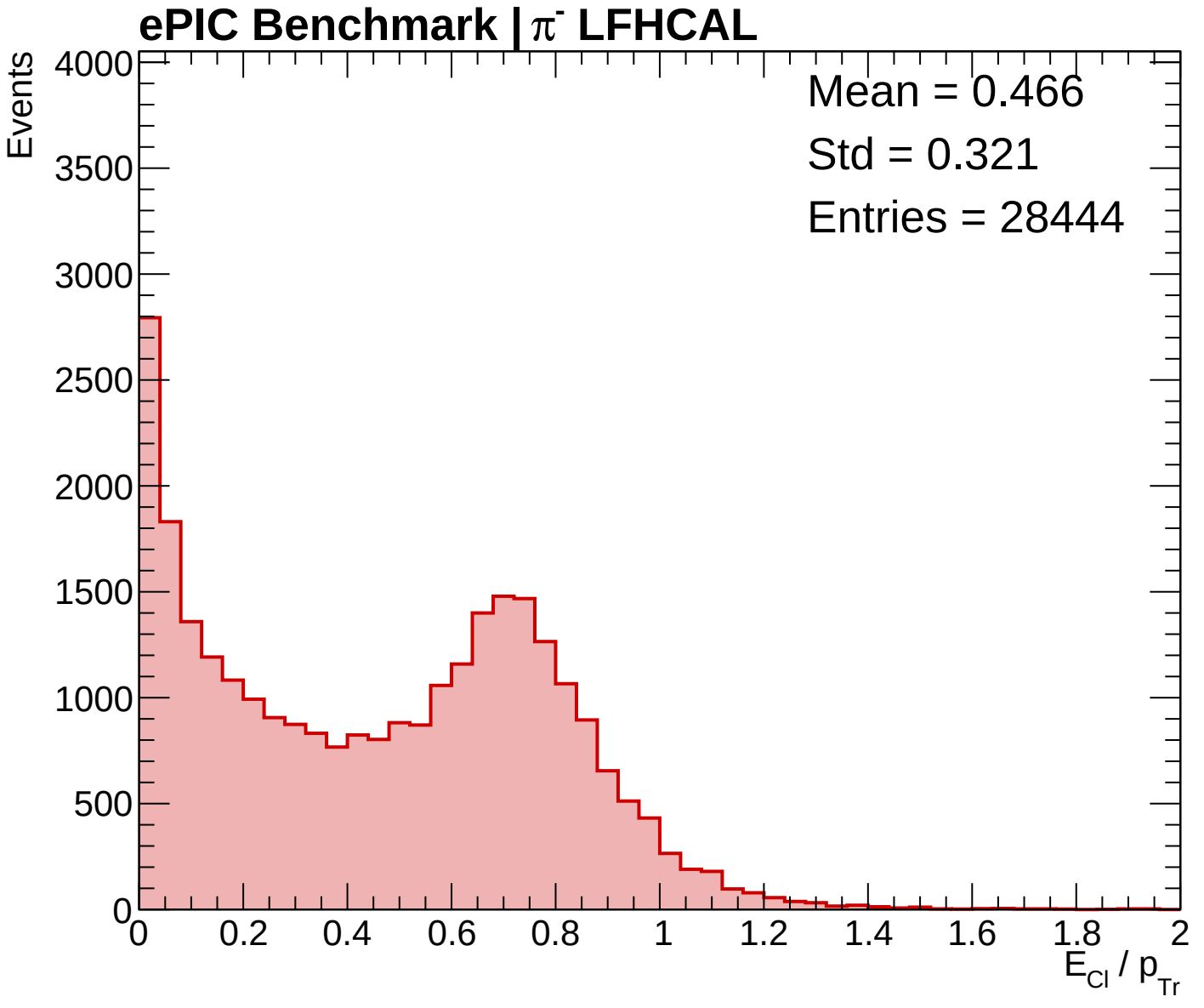

ePIC Benchmark | π⁻ LFHCAL
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
1
1.5
2
2.5
3
3.5
η
MC truth
MC matched
Matched: 50.7 %
1.0 < η < 3.5

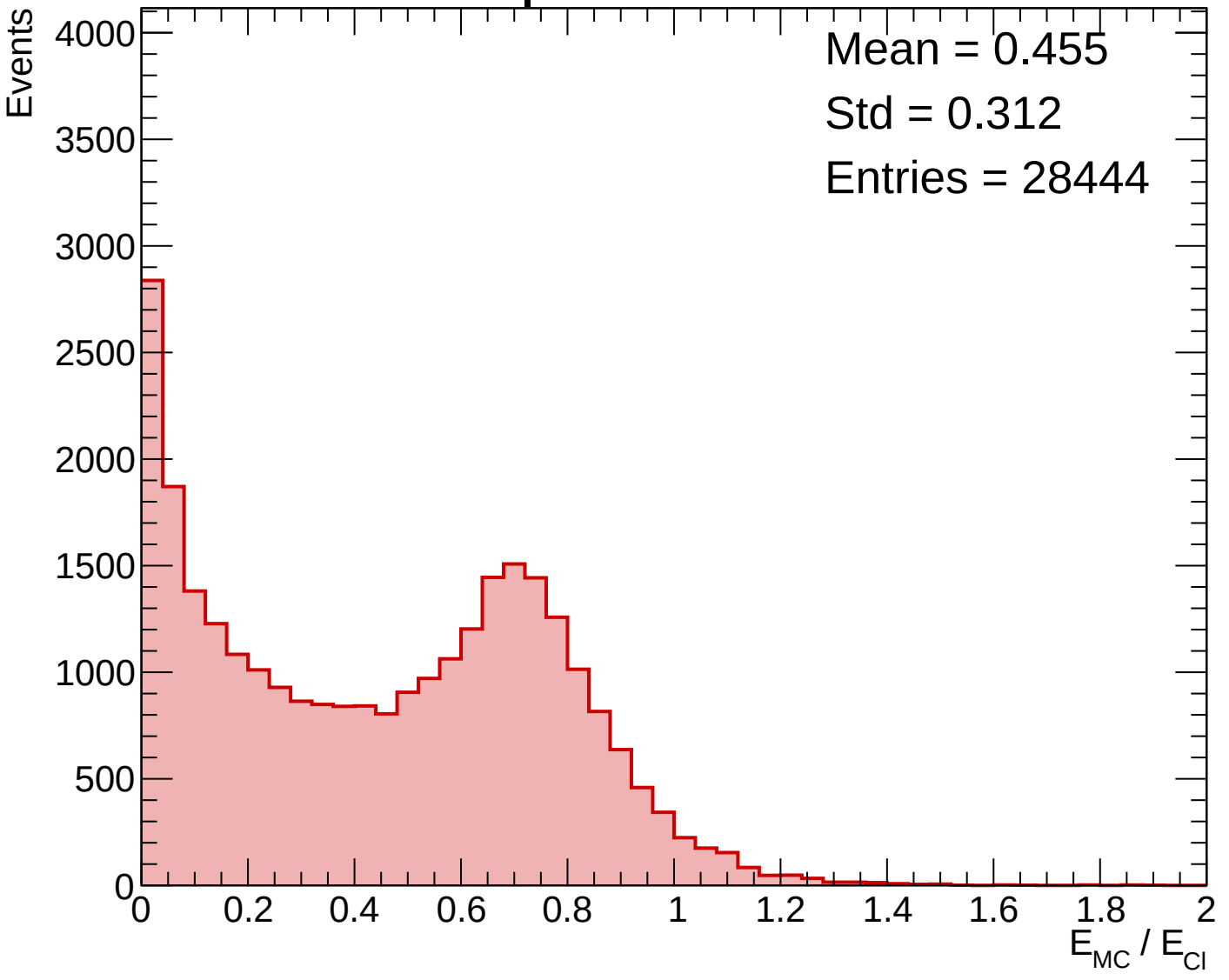

ePIC Benchmark | π⁻ LFHCAL
Particles
7000
6000
5000
4000
3000
2000
1000
0
0
2
4
6
8
10
12
p_T [GeV]
MC truth
MC matched
Matched: 50.7 %
1.0 < η < 3.5

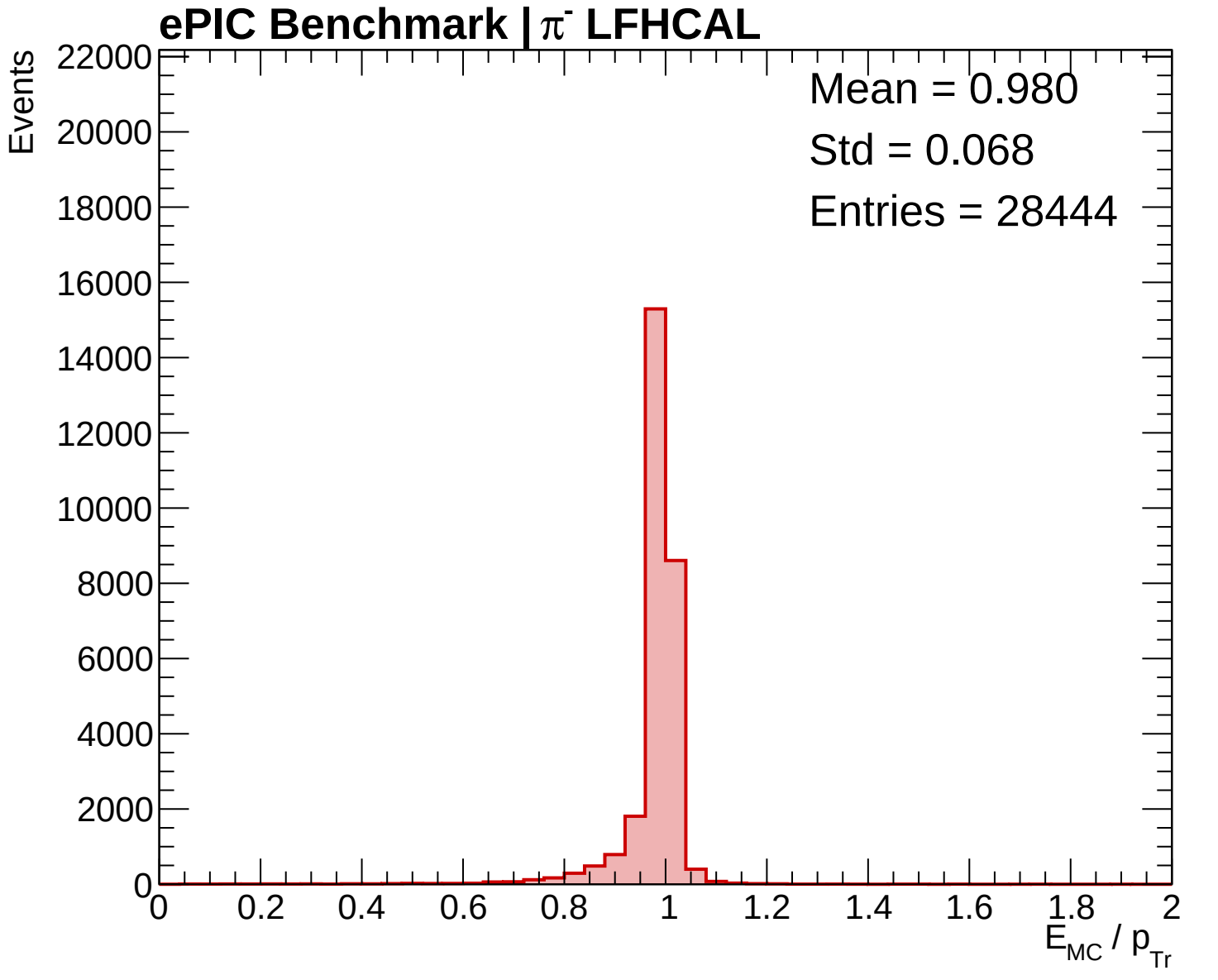

ePIC Benchmark | π⁻ LFHCAL
Particles
1800
1600
1400
1200
1000
800
600
400
200
0
–3
–2
–1
0
1
2
3
ϕ [rad]
MC truth
MC matched
Matched: 50.7 %
1.0 < η < 3.5

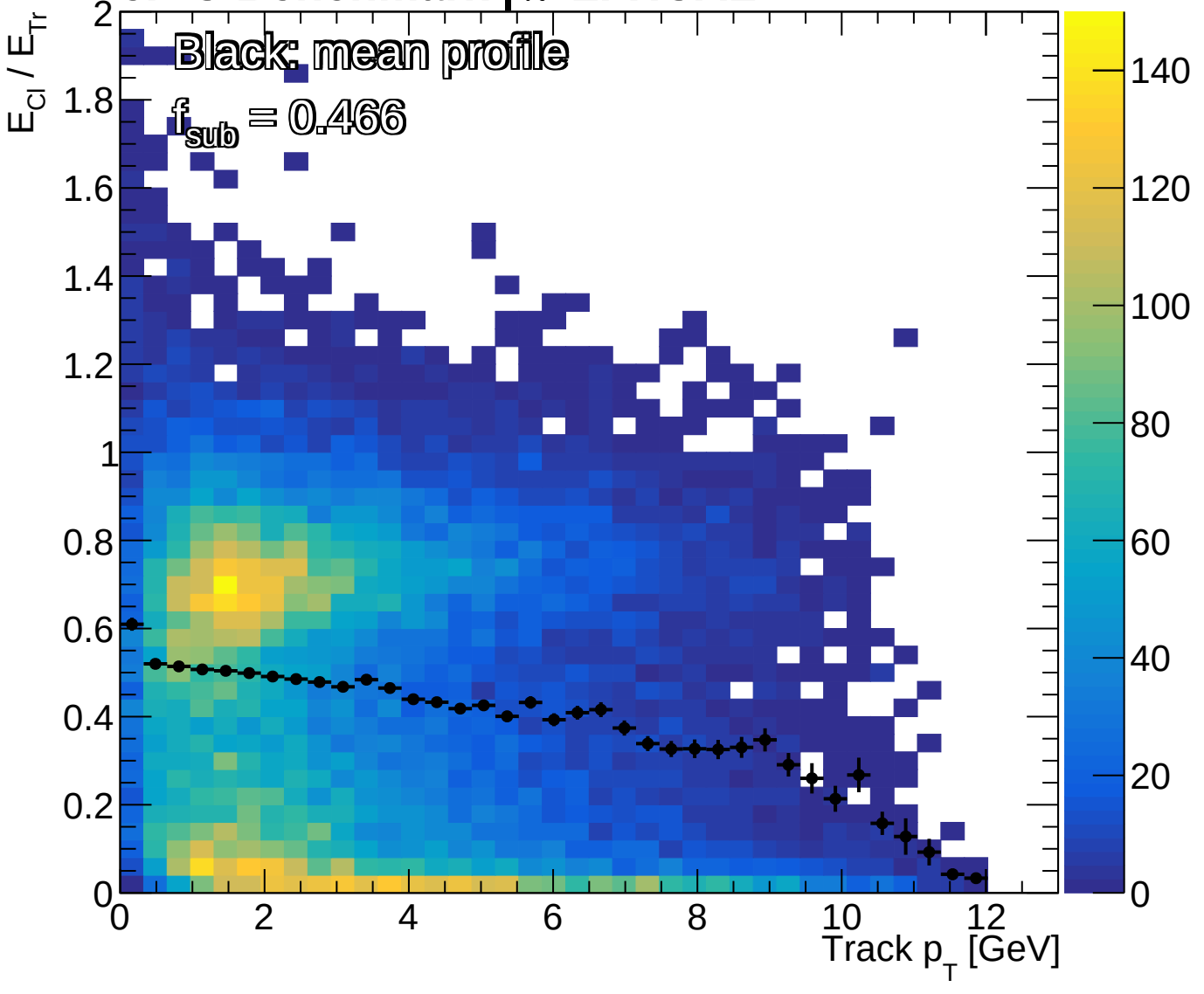

ePIC Benchmark | π⁻ LFHCAL
Δϕ [rad]
0.2
0.15
0.1
0.05
0
–0.05
–0.1
–0.15
–0.2
–0.2
–0.15
–0.1
–0.05
0
0.05
0.1
0.15
0.2
Δη
220
200
180
160
140
120
100
80
60
40
20
0

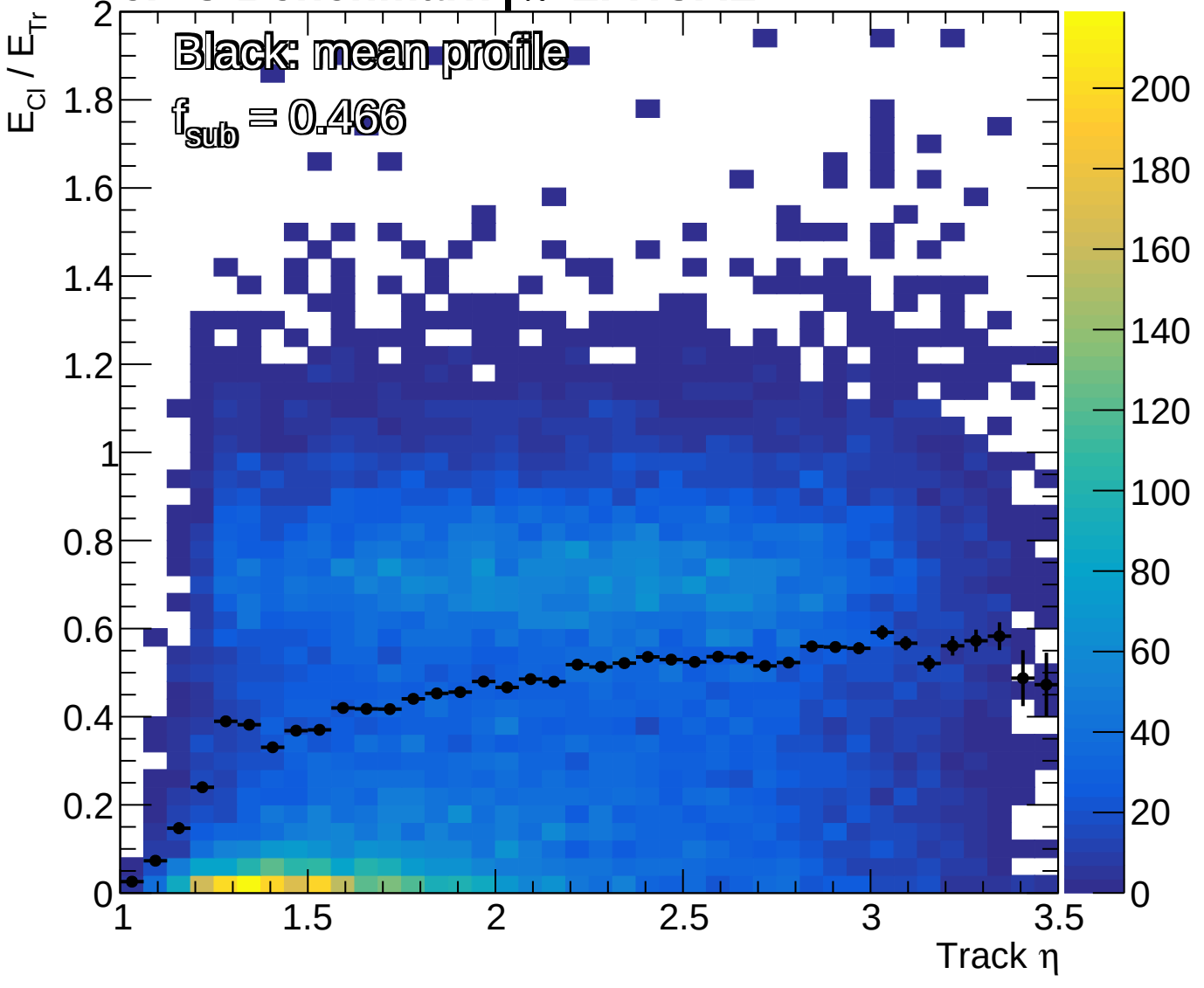

ePIC Benchmark | π- LFHCAL
Clusters
7000
6000
5000
4000
3000
2000
1000
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
0.5
$R_{\eta\phi} = \sqrt{\Delta\eta^2+\Delta\phi^2}$

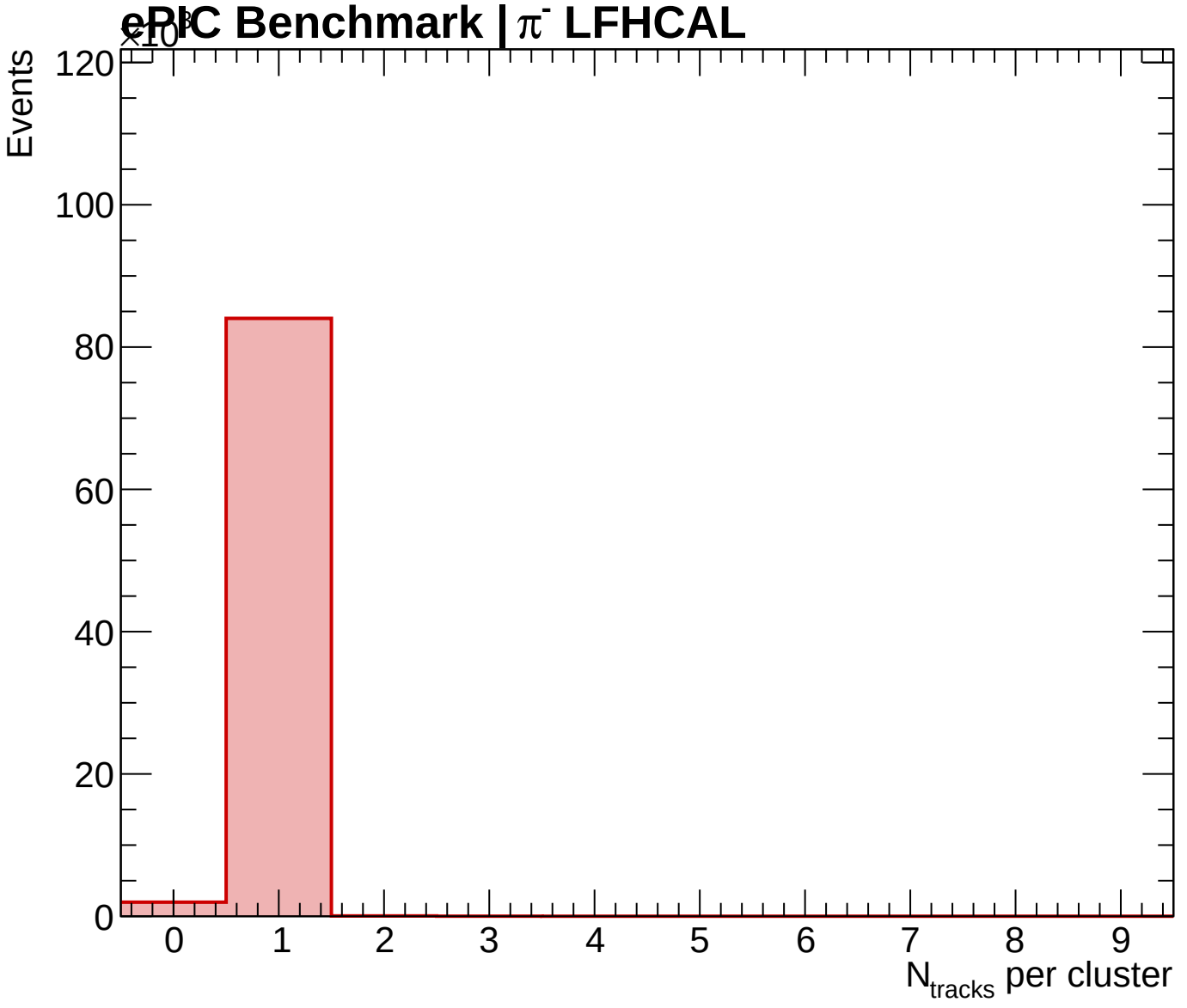

ePIC Benchmark | π⁻ LFHCAL
Mean = 0.466
Std = 0.321
Entries = 28444
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_Cl / p_Tr

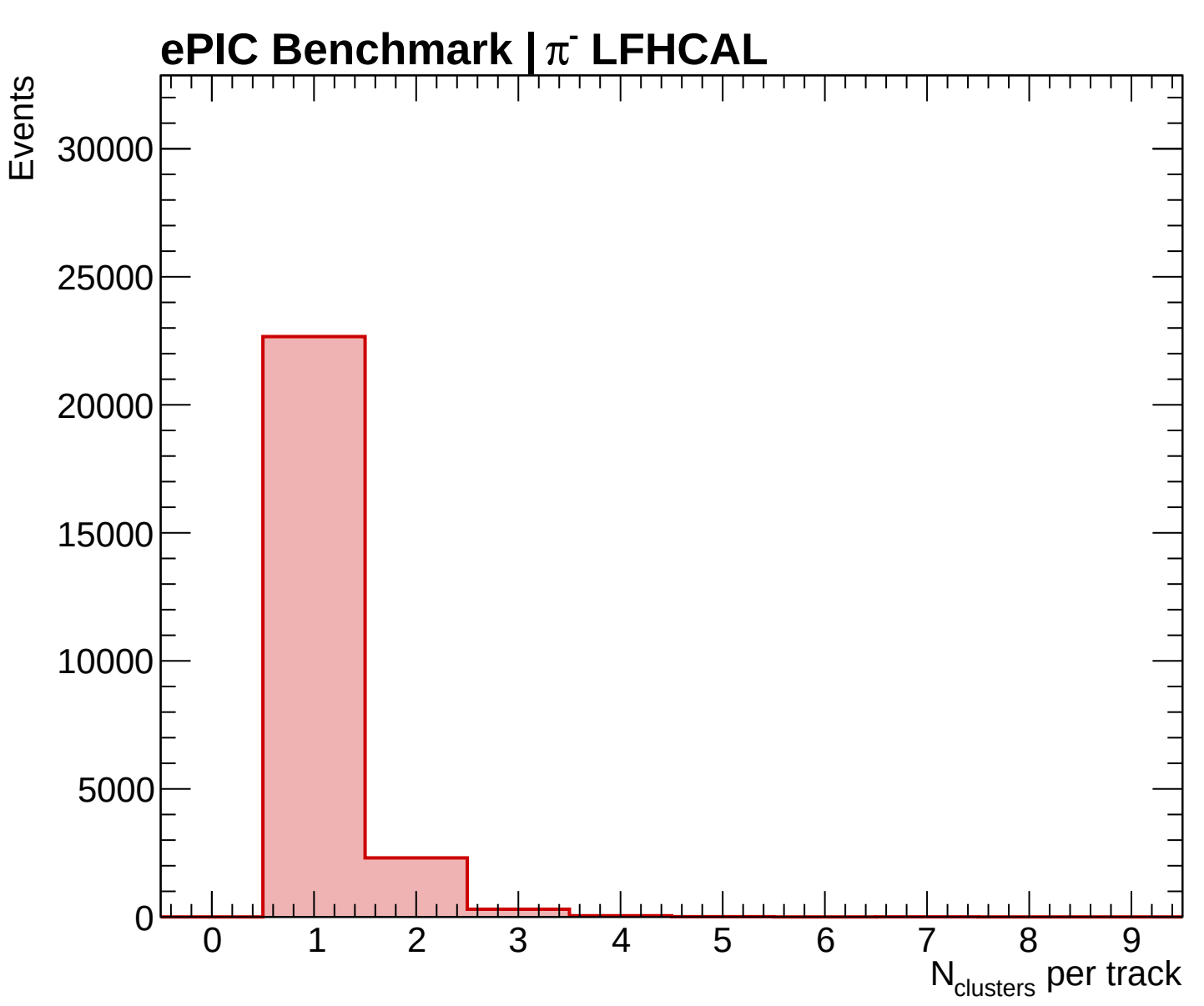

ePIC Benchmark | π⁻ LFHCAL
Mean = 0.455
Std = 0.312
Entries = 28444
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / E_Cl

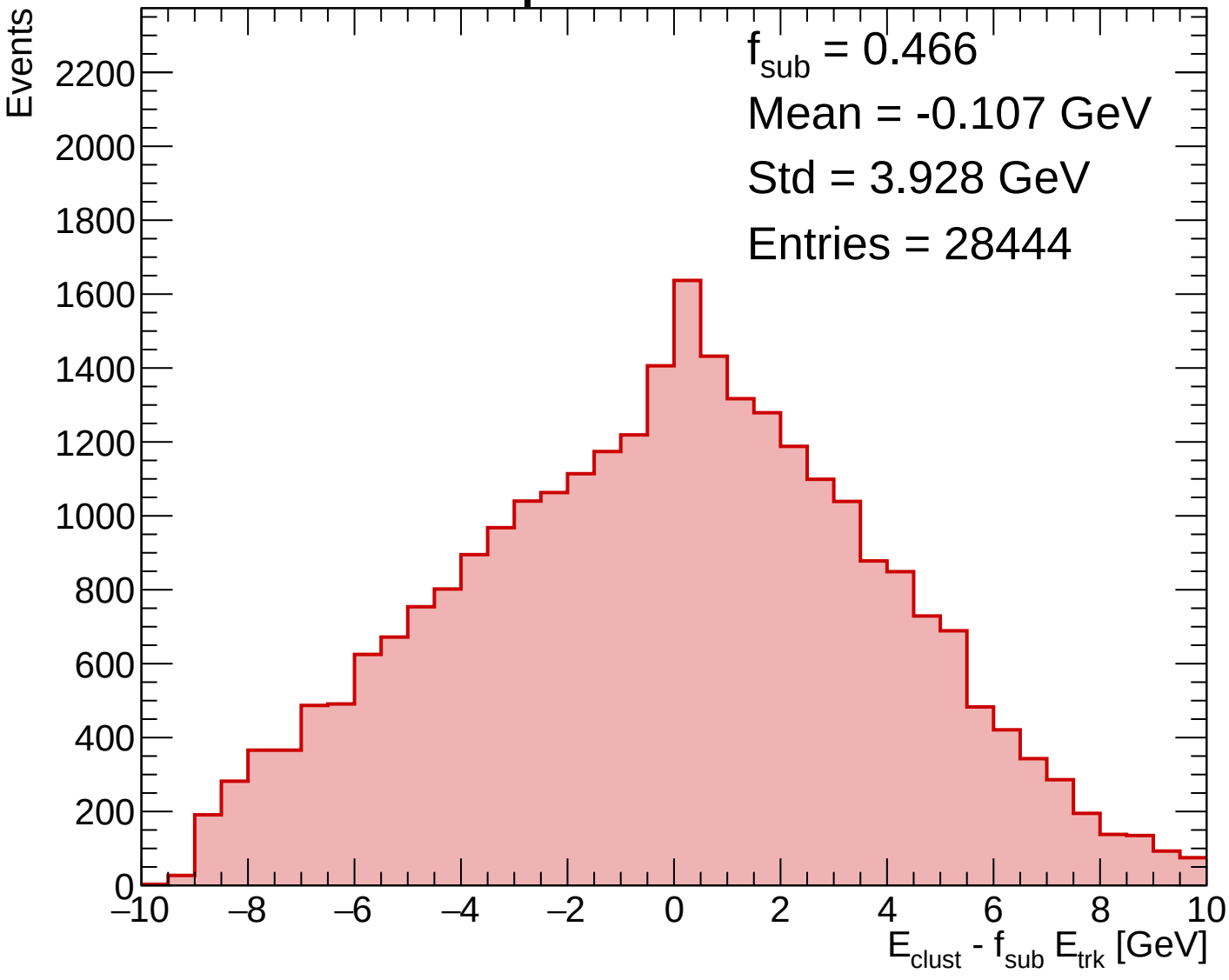

ePIC Benchmark | π⁻ LFHCAL
Mean = 0.980
Std = 0.068
Entries = 28444
Events
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
E_MC / p_Tr

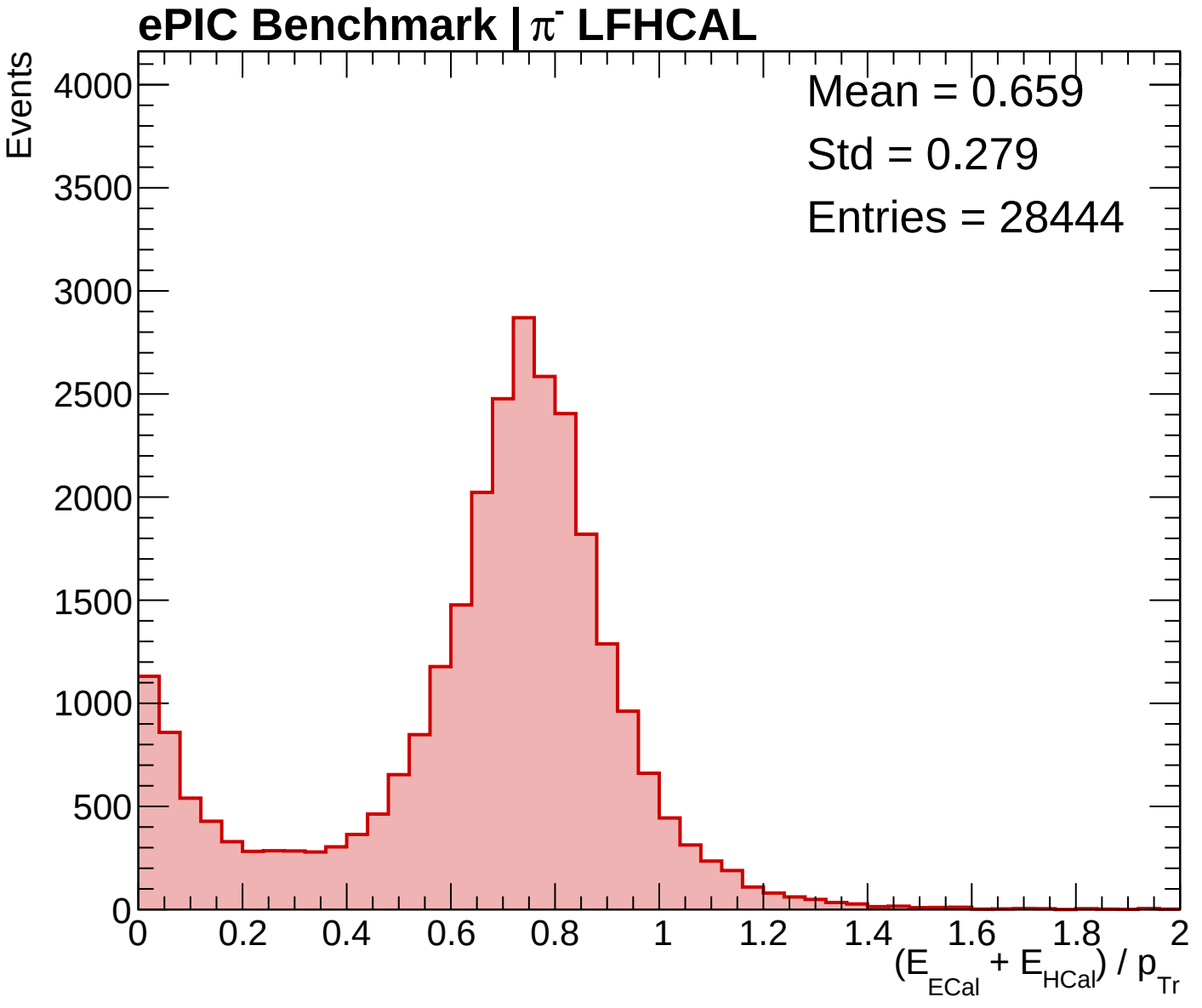

ePIC Benchmark | π⁻ LFHCAL
Black: mean profile
f_sub = 0.466
E_Cl / E_Tr
Track p_T [GeV]
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0
2
4
6
8
10
12
0
20
40
60
80
100
120
140


ePIC Benchmark | π⁻ LFHCAL
Black: mean profile
f_sub = 0.466
E_Cl / E_Tr
Track η
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
1
1.5
2
2.5
3
3.5
0
20
40
60
80
100
120
140
160
180
200


ePIC Benchmark | π- LFHCAL
×10³
Events
120
100
80
60
40
20
0
0
1
2
3
4
5
6
7
8
9
N_tracks per cluster


ePIC Benchmark | π- LFHCAL
Events
30000
25000
20000
15000
10000
5000
0
0
1
2
3
4
5
6
7
8
9
N clusters per track


ePIC Benchmark | π⁻ LFHCAL
Events
f_sub = 0.466
Mean = -0.107 GeV
Std = 3.928 GeV
Entries = 28444
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
–10
–8
–6
–4
–2
0
2
4
6
8
10
E_clust - f_sub E_trk [GeV]


ePIC Benchmark | π⁻ LFHCAL
Mean = 0.659
Std = 0.279
Entries = 28444
Events
4000
3500
3000
2500
2000
1500
1000
500
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
(E_ECal + E_HCal) / p_Tr